Zexu Chen

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Summary

PhD candidate (expected May 2028) in computational materials science at Ohio State, combining phase-field simulation, GPU-accelerated HPC, and machine learning to accelerate alloy design. Two peer-reviewed publications on NiTi shape memory alloys. Seeking 2026 summer R&D internship in materials informatics, computational alloy design, or ML for materials.

Education

The Ohio State University — PhD, Materials Science and Engineering May 2024 – May 2028 (expected)

Advisor: Prof. Yunzhi Wang

The Ohio State University — MS, Materials Science and Engineering Aug 2022 – May 2024

Sichuan University / The Ohio State University — BSc, Materials Science Aug 2017 – May 2022

Publications

- Sriram H., Feng L., **Chen Z.**, Wang Y. (2026). Soft and Hard Confinement Effects on Martensitic Transformation in NiTi SMAs: Insights from Phase-Field Simulations. *Shape Memory and Superelasticity*. https://doi.org/10.1007/s40830-026-00619-3
- Abe O., Qiu Z., **Chen Z.**, Jinschek J.R., Gouma P.-I. (2021). Effect of Crystallite Size on Low-Temperature Solid-Solid Phase Transformations in WO_3. *Ceramics International*. https://doi.org/10.1016/j.ceramint.2021.08.254

Research Experience

Graduate Researcher, Wang Group, The Ohio State University *Aug 2022 – Present*

- Apply Ni_4Ti_3 precipitation field to GPU-accelerated 3D phase-field framework (Allen-Cahn / Cahn-Hilliard, OpenACC, SLURM on OSC) for martensitic transformation in NiTi alloys.
- Demonstrated linear superelastic response via concentration modulation and amorphous-crystalline composite microstructures; published in *Shape Memory and Superelasticity*.
- Building ML surrogates (Gaussian Process Regression, LASSO, Bayesian optimization on CALPHAD-informed datasets) to map composition and microstructure to transformation temperature and stress-strain response for inverse alloy design.

ML for Superconductivity Prediction, MATSCEN 7193.02 (Machine Learning for Materials Research) *Spring 2026*

- Built XGBoost and GPR pipeline to predict superconducting parameters (λ, T_c) from Eliashberg spectral functions $\alpha^2F(\omega)$, using physics-informed features (frequency-weighted α^2F/ω bins) aligned with the Eliashberg integral.
- Tuned GPR with Matérn ($\nu = 1.5/2.5$) + White kernels to capture spectral discontinuities, and benchmarked bin resolution (up to 800) vs. generalization to characterize the curse-of-dimensionality trade-off.

Teaching

- Graduate Teaching Assistant, OSU (Oct 2023 – Oct 2024): *Introduction to Engineering Materials*; *Modeling and Simulation Lab II*.

Conference Presentations

- TMS 2026 (poster), MS&T 2025 (oral), TMS 2025 (poster), MS&T 2023 (oral) — phase-field modeling of structural/compositional inhomogeneous NiTi shape memory alloys.

Professional Service

- Reviewer, *Shape Memory and Superelasticity* (ASM International / Springer), 2026 – Present.

Technical Skills

Programming: Python (NumPy, SciPy, scikit-learn, PyTorch), C, MATLAB, Bash
HPC / GPU: OpenACC, CUDA, SLURM, Linux, Git, Ohio Supercomputer Center
Simulation: Phase-field (Allen-Cahn, Cahn-Hilliard, multiphase-field), Khachaturyan microelasticity, CALPHAD, LAMMPS, COMSOL, OVITO, ParaView
Machine Learning: LASSO, Bayesian optimization, Gaussian Process Regression, Physics-Informed Neural Networks
Domain: Shape memory alloys (NiTi), martensitic transformation, ICME, materials informatics
Languages: English (fluent), Mandarin (native)